



COLORADO
Division of Reclamation,
Mining and Safety
Department of Natural Resources

November 21, 2024

Robin Simons
CEMEX, Inc.
P.O. Box 529
Lyons, CO 80540

RE: Division's Financial Warranty Cost Estimates for Lyons Quarry (M1977-208)

Dear Robin Simons,

On March 18, 2024, the Division of Reclamation Mining and Safety (Division/DRMS) received CEMEX's responses to the Division's letter dated February 16, 2024. The Division used those responses, along with previous responses, and permit file information to update the financial warranty for the Lyons Quarry.

A copy of the mine's estimate is attached for your review. The Division will allow 30 days or until **December 21, 2024** to review and comment on the bond estimate prior to issuing a surety increase notice for the Lyons Quarry. Once the surety increase is issued by the Division, the Operator will have 60 days to provide the additional financial warranty.

If you need additional information or have any questions, please contact me by email at patrick.lennberg@state.co.us.

Sincerely,

Patrick Lennberg
Environmental Protection Specialist

cc: Jared Ebert, DRMS

ec: Robin Simons, CEMEX, Inc., robing.simons@cemex.com



Attachments

COST SUMMARY WORK

Task description: Cost Summary

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 000

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-000

User: JPL

Agency or organization name: DRMS

TASK LIST (DIRECT COSTS)

Task	Description	Form Used	Fleet Size	Task Hours	Cost
001	Table 2 Demolition	DEMOLISH	1	22,600.00	\$11,811,459
002	Table 3 Steel Demo	DEMOLISH	1	615.00	\$779,767
003	Table 5 Other Reclamation Costs	DEMOLISH	1	2,700.00	\$410,681
004	Excavation of On-Site Disposal Cell	SCRAPER1	1	83.35	\$384,267
005	Installation of GCL	NA	1	54.20	\$174,000
006	3 Ft Clay Liner Protective Cover	SCRAPER1	1	23.22	\$101,833
007	Interstitial Fill	SCRAPER1	1	14.89	\$68,669
008	30-inch Cover	SCRAPER1	1	22.07	\$101,732
009	Stormwater Diversion Ditch	POSTMININ G	1	322.83	\$86,204
010	B-Pit Topsoil Stripping	DOZER	1	25.66	\$12,247
011	Backfill C Pit	TRUCK1	2	111.36	\$380,544
012	Shale from SE Corner of site to C-Pit	SCRAPER1	1	32.02	\$151,159
013	Rough Grade Plant Area	GRADER	1	6.26	\$2,265
014	Fine Grade Plant Area	GRADER	1	13.57	\$4,915
015	Rough Grade C-Pit	GRADER	1	5.75	\$2,082
016	Fine Grade C-Pit	GRADER	1	17.91	\$6,488
017	Rip Roads and Facilities	RIPPER	1	5.41	\$2,629
018	Grade Roads and Facilities	GRADER	1	1.74	\$632
019	Rip Disposal Cell	RIPPER	2	8.99	\$8,742
020	Grade Disposal Cell	GRADER	1	6.14	\$2,226
021	Grade B-Pit Borrow	GRADER	1	9.70	\$3,514
022	Grade Topsoil Stockpile	GRADER	1	4.02	\$1,456
023	Topsoil Plant Area	SCRAPER1	1	26.84	\$117,734
024	Topsoil C-Pit	SCRAPER1	1	23.30	\$102,209
025	Topsoil Roads and Facilities	SCRAPER1	1	2.24	\$9,806
026	Topsoil Disposal Cell	SCRAPER1	1	7.43	\$32,600
027	Topsoil B-Pit Borrow	DOZER	2	19.25	\$18,353
028	Plant Area Revegetation	REVEGE	1	53.00	\$69,065
029	On-Site Disposal Cell Revegetation	REVEGE	1	26.00	\$34,200
030	C-Pit Area Revegetation	REVEGE	1	46.00	\$59,960
031	Roads and Facilities Revegetation	REVEGE	1	7.00	\$8,883
032	B-Pit Borrow Area	REVEGE	1	36.00	\$46,636
033	Topsoil Stockpile Revegetation	REVEGE	1	15.00	\$19,321
034	Weed Control	REVEGE	1	15.00	\$2,914
035	C-Pit Water Treatment TR-18	NA	1	1.00	\$5,752,000
036	Well Abandonment	BOREHOLE	1	16.00	\$6,207
037	Mob/Demob	MOBILIZE	1	5.06	\$90,136

<u>SUBTOTALS:</u>	26983.21	\$20,867,535
--------------------------	-----------------	---------------------

INDIRECT COSTS**OVERHEAD AND PROFIT:**

Liability insurance:	2.02	Total =	\$421,524
Performance bond:	1.05	Total =	\$219,109
Job superintendent:	3,126.00	Total =	\$247,798
Profit:	10.00	Total =	\$2,086,754
		TOTAL O & P =	\$2,975,185
		CONTRACT AMOUNT (direct + O & P) =	\$23,842,720

LEGAL - ENGINEERING - PROJECT MANAGEMENT:

Financial warranty processing (legal/related costs):	\$500	Total =	\$500
Engineering work and/or contract/bid preparation:	4.00	Total =	\$953,709
Reclamation management and/or administration:	5.00		\$1,192,136

CONTINGENCY:	3.00	Total =	\$626,026
--------------	------	---------	-----------

TOTAL INDIRECT COST =	\$5,747,556
-----------------------	-------------

TOTAL BOND AMOUNT (direct + indirect) =	\$26,615,091
--	---------------------

DEMOLITION WORK

Task description:

Table 2 DemolitionSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 001State: ColoradoAbbreviation: NoneDate: 2/13/2024County: BoulderFilename: M208-001User: JPLAgency or organization name: DRMS**UNIT COSTS****Location adjustment: 91.80 %**

Structure or Item Description	Dimensions	Demolition Menu Selection	Quantity	Unit	Unit Cost	Total Cost
A-Admin Building	105x46x14	Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	67,620.00	CF	\$0.29	\$19,454.27
A-Admin Bld floor	105x46x0.5	Demo. and on-site disposal in existing pit, 6 in. thick - Max. 10,000 ft. haul	4,830.00	SF	\$1.16	\$5,585.90
B-Mill Bld	168x97x53	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	863,688.00	CF	\$1.10	\$954,116.13
B-Mill Bld Floor	168x97x0.5	Demo. and on-site disposal in existing pit, 6 in. thick - Max. 10,000 ft. haul	16,296.00	SF	\$1.16	\$18,846.32
B-Raw Mill Foundation	Sum of LF Provided	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	107.00	LF	\$13.88	\$1,484.99
B-Finish Mill Foundation	Sum of LF Provided	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	168.00	LF	\$13.88	\$2,331.57
B-Foundation	Sum of LF Provided	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	225.00	LF	\$13.88	\$3,122.64
C-Control Center	88x37x38	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	123,728.00	CF	\$1.10	\$136,682.32
C-Control Center b	37x34x20	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	25,160.00	CF	\$1.10	\$27,794.25
C-Control Center Floor	88x37	Demo. and on-site disposal in existing pit, 6 in. thick - Max. 10,000 ft. haul	3,265.00	SF	\$1.16	\$3,775.97
D-Raw Material Silos	0.58x73x27	Explosive demolition, large projects - Concrete structures	38,263.00	CF	\$0.41	\$15,687.83

D-Raw Material Silo Debris Disp	785	Explosive demo, Disposal of material average	785.00	CY	\$14.75	\$11,578.75
E-Pack House a	150x72x25	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	270,000.00	CF	\$1.10	\$298,269.00
E-Pack House b	45x41x36	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	66,420.00	CF	\$1.10	\$73,374.17
E-Pack House a floor	150x72x1	Demo. and on-site disposal in existing pit, 12 in. thick - Max. 10,000 ft. haul	7,560.00	SF	\$2.31	\$17,487.04
E-Pack House b Floor	45x41x1	Demo. and on-site disposal in existing pit, 12 in. thick - Max. 10,000 ft. haul	1,845.00	SF	\$2.31	\$4,267.67
F-Clinker Storage Building	388x200x79	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	6,130,400.00	CF	\$1.10	\$6,772,252.88
F-Clinker Silo	0.58x73x27	Explosive demolition, large projects - Concrete structures	573.00	CF	\$0.41	\$234.93
F-Clinker Silo Disp	0.58x73x27	Explosive demo, Disposal of material average	21.00	CY	\$14.75	\$309.75
F-Clinker Floor	388x200	Demo. and on-site disposal in existing pit, 12 in. thick - Max. 10,000 ft. haul	77,600.00	SF	\$2.31	\$179,496.56
G-Finished Cement Silos (15)	589503 total	Explosive demolition, large projects - Concrete structures	589,503.00	CF	\$0.41	\$241,696.23
G-Finish Silo Disposal	589503	Explosive demo, Disposal of material average	21,833.00	CY	\$14.75	\$322,036.75
H Pump House	25x25x15	Bldg. (SC) demo./on- site disposal in existing pit or cut - Max. 10,000 ft. haul	9,375.00	CF	\$0.29	\$2,697.19
H-Pump House Floor	25x15x0.67	Demo. and on-site disposal in existing pit, 6 in. thick - Max. 10,000 ft. haul	625.00	SF	\$1.16	\$722.81
J-Electrical Room	26x15x15	Bldg. (SC) demo./on- site disposal in existing pit or cut - Max. 10,000 ft. haul	5,850.00	CF	\$0.29	\$1,683.05
J-Electrical Room Floor	26x15	Demo. and on-site disposal in existing pit, 6 in. thick - Max. 10,000 ft. haul	390.00	SF	\$1.16	\$451.04
T-Dust Suppressor	70x30x60	Plant (3C) demo./off- site disposal in approved landfill - Max. 15 mile haul	126,000.00	CF	\$1.03	\$130,258.80

V-Coal Unloading Building	71x36x30	Plant (3S) demo./off-site disposal in approved landfill - Max. 15 mile haul	76,680.00	CF	\$0.88	\$67,225.36
V-Coal Unloading Bases	71x36	Demo. and on-site disposal in existing pit, 8 in. thick - Max. 10,000 ft. haul	2,556.00	SF	\$1.54	\$3,941.35
AA-Maintenance Shop	120x65x24	Plant (3S) demo./off-site disposal in approved landfill - Max. 15 mile haul	187,200.00	CF	\$0.88	\$164,118.24
AA-Maintenance Shop Floor	120x65	Demo. and on-site disposal in existing pit, 10 in. thick - Max. 10,000 ft. haul	7,800.00	SF	\$1.93	\$15,035.28
I-Wash House	10x10x8	Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	800.00	CF	\$0.29	\$230.16
I-Wash House Floor	10x10	Demo. and on-site disposal in existing pit, 8 in. thick - Max. 10,000 ft. haul	100.00	SF	\$1.54	\$154.20
K-Mobile Equipmt Shed	103x64x35	Bldg. (SN) demo./off-site disposal in approved landfill - Max. 15 mile haul	230,720.00	CF	\$0.42	\$97,110.05
K-Mobile Equipment Shed Floor	103x64x0.67	Demo. and on-site disposal in existing pit, 8 in. thick - Max. 10,000 ft. haul	6,592.00	SF	\$1.54	\$10,164.86
L-Burner Bld a	120x91x53	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	578,760.00	CF	\$1.10	\$639,356.17
L-Burner Bld b	48x91x25	Plant (1C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	109,200.00	CF	\$1.04	\$113,458.80
L-Burner Bld Foundation	700 LF	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	700.00	LF	\$13.88	\$9,714.88
M-Primary Crusher - Steel	10x10x10	Plant (1S) demo./off-site disposal in approved landfill - Max. 15 mile haul	100.00	CF	\$0.85	\$84.86
M-Primary Crusher Concrete	40x20x30	Plant (3C) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	25,800.00	CF	\$1.10	\$28,501.26
M-Primary Crusher Floor	10x10x067	Demo. and on-site disposal in existing pit, 8 in. thick - Max. 10,000 ft. haul	100.00	SF	\$1.54	\$154.20
N-Secondary Crusher Silos	97993 cf total	Explosive demolition, large projects - Steel structures	97,993.00	CF	\$0.41	\$40,177.13

N-Secondary Crusher Silos Disposal	3629 cy	Loading and 5 mile haul, salvage allowed - Steel frame structures	3,629.00	CY	\$13.75	\$49,898.75
N-Secondary Crusher Floor	1x32x1	Demo. and on-site disposal in existing pit, 12 in. thick - Max. 10,000 ft. haul	32.00	SF	\$2.31	\$74.02
O-Homogenizing Silos (3)	256739 cf	Explosive demolition, large projects - Concrete structures	256,739.00	CF	\$0.41	\$105,262.99
O-Silo Disposal	9508 cy	Explosive demo, Disposal of material average	9,508.00	CY	\$14.75	\$140,243.00
P-Coal Facilities	261383 cf	Explosive demolition, large projects - Concrete structures	261,383.00	CF	\$0.41	\$107,167.03
P-Coal Facilities Disposal	9680 cy	Explosive demo, Disposal of material average	9,680.00	CY	\$14.75	\$142,780.00
P-Coal Facilities Foundation	10x10x5	Demo. and on-site disposal in existing pit, 12 in. thick - Max. 10,000 ft. haul	100.00	SF	\$2.31	\$231.31
Q-Stack	1x195x44	Explosive demolition, large projects - Concrete structures	270,161.00	CF	\$0.41	\$110,766.01
Q-Stack Disposal	976 cy	Explosive demo, Disposal of material average	976.00	CY	\$14.75	\$14,396.00
R-Bag House	130x60x45	Plant (3S) demo./off-site disposal in approved landfill - Max. 15 mile haul	351,000.00	CF	\$0.88	\$307,721.70
S-Raw Material Dryer	21x16x16	Plant (1S) demo./off-site disposal in approved landfill - Max. 15 mile haul	5,376.00	CF	\$0.85	\$4,562.07
S-Raw Material Dryer Foundation	Variable	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	74.00	LF	\$13.88	\$1,027.00
U-Warehouse	122x74.5x20	Bldg. (SN) demo./off-site disposal in approved landfill - Max. 15 mile haul	181,780.00	CF	\$0.42	\$76,511.20
U-Warehouse Foundation	122 LF	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	392.00	LF	\$13.88	\$5,440.33
X-Brick Shed	96x40x18	Bldg. (SC) demo./on-site disposal in existing pit or cut - Max. 10,000 ft. haul	69,120.00	CF	\$0.29	\$19,885.82
X-Brick Shed Floor	96x40	Demo. and on-site disposal in existing pit, 8 in. thick - Max. 10,000 ft. haul	3,840.00	SF	\$1.54	\$5,921.28
Z-Kiln	10x24x11	Plant (1S) demo./off-site disposal in approved	2,640.00	CF	\$0.85	\$2,240.30

		landfill - Max. 15 mile haul				
BB-PreHeater Tower	55x40x60	Explosive demolition, large projects - Steel structures	132,000.00	CF	\$0.41	\$54,120.00
BB-PreHeater Tower Disposal	4888 cy	Loading and 5 mile haul, salvage allowed - Steel frame structures	4,888.00	CY	\$13.75	\$67,210.00
BB-PreHeater Foundation	55 LF	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	190.00	LF	\$13.88	\$2,636.90
CC-Oil House	16x32x11	Bldg. (SN) demo./off-site disposal in approved landfill - Max. 15 mile haul	5,632.00	CF	\$0.42	\$2,370.51
CC-Oil House Foundation	96 LF	Demo. and on-site disposal in existing pit, 1.5 ft. x 2 ft. - Max. 10,000 ft. haul	96.00	LF	\$6.94	\$666.16
DD-Blower Building	32x12x10	Plant (1S) demo./off-site disposal in approved landfill - Max. 15 mile haul	3,840.00	CF	\$0.85	\$3,258.62
DD-Blower Building Foundation	88 LF	Demo. and on-site disposal in existing pit, 1.5 ft. x 2 ft. - Max. 10,000 ft. haul	88.00	LF	\$6.94	\$610.65
EE-SNCR	10x12x11	Plant (1S) demo./off-site disposal in approved landfill - Max. 15 mile haul	1,320.00	CF	\$0.85	\$1,120.15
EE-SNCR Foundation	44 LF	Demo. and on-site disposal in existing pit, 2.0 ft. x 3 ft. - Max. 10,000 ft. haul	44.00	LF	\$13.88	\$610.65
Pavement-All outside Pavement	22,222 cy	USER PROVIDED ITEM	22,222.00	CY	\$54.21	\$1,204,654.62

		Subtotal	Total Cost
Job Hours:	<u>22,600.00</u>	(unadjusted):	(adjusted for location):
		<u>\$12,866,512.68</u>	<u>\$11,811,458.64</u>

DEMOLITION WORK

Task description: **Table 3 Steel Demo**

Site: **Lyons Quarry**

Permit Action: **2024 Cost Est**

Permit/Job#: **M1977208**

PROJECT IDENTIFICATION

Task #: **002**
Date: **2/14/2024**
User: **JPL**

State: **Colorado**
County: **Boulder**

Abbreviation: **None**
Filename: **M208-002**

Agency or organization name: **DRMS**

UNIT COSTS

Location adjustment: 91.80 %

Structure or Item Description	Dimensions	Demolition Menu Selection	Quantity	Unit	Unit Cost	Total Cost
B-Mill Building Hoppers	15x27x27	USER PROVIDED ITEM	231,885.00	CF	\$1.24	\$287,537.40
B-Mill Building Hoppers	12x20x20	USER PROVIDED ITEM	75,396.00	CF	\$1.24	\$93,491.04
B-Mill Building Hopper	4x40x40	USER PROVIDED ITEM	201,060.00	CF	\$1.24	\$249,314.40
B-Mill Building Hopper	1x30x30	USER PROVIDED ITEM	21,206.00	CF	\$1.24	\$26,295.44
B-Mill Building Hopper	1x26x26	USER PROVIDED ITEM	13,804.00	CF	\$1.24	\$17,116.96
B-Mill Building Hopper	1x32x32	USER PROVIDED ITEM	25,736.00	CF	\$1.24	\$31,912.64
B-Mill Building Hopper	1x21x25.2	USER PROVIDED ITEM	8,728.00	cf	\$1.24	\$10,822.72
B-Mill Building Hopper	1x20x20	USER PROVIDED ITEM	6,283.00	CF	\$1.24	\$7,790.92
B-Mill Building Stairs	1x3x300	USER PROVIDED ITEM	900.00	CF	\$1.06	\$954.00
D-Raw Mat. Silos Stairs	Various	USER PROVIDED ITEM	6,270.00	CF	\$1.06	\$6,646.20
D-Raw Mat Fencing	800 sf	Fencing, chain link, including posts and fabric - 8 ft. to 10 ft. high	100.00	LF	\$3.53	\$353.00
D-Raw Mat Conveyors	287 cy	Conveyor, demolition, off-site disposal in approved landfill, 15 mile haul	7,506.00	CF	\$0.85	\$6,380.10
E-Pack House Stairs	Various	USER PROVIDED ITEM	6,750.00	CF	\$1.06	\$7,155.00
F-Clinker Storage Bld Stairs	Various	USER PROVIDED ITEM	1,500.00	CF	\$1.06	\$1,590.00
F-Clinker Storage Bld Conveyors	167 cy	Conveyor, demolition, off-site disposal in approved landfill, 15 mile haul	4,509.00	CF	\$0.85	\$3,832.65
G-Finish Cement Stairs	Various	USER PROVIDED ITEM	2,100.00	CF	\$1.06	\$2,226.00
G-Finish Cement Fencing	300 LF	Fencing, chain link, including posts and fabric - 8 ft. to 10 ft. high	300.00	LF	\$3.53	\$1,059.00
G-Finish Cement Grating	16587 sf	USER PROVIDED ITEM	16,578.00	SF	\$1.06	\$17,572.68

T-Dust Suppressor Stairs	750 CF	USER PROVIDED ITEM	750.00	CF	\$1.06	\$795.00
W-Power Substation Fence	440	Fencing, chain link, including posts and fabric - 8 ft. to 10 ft. high	440.00	LF	\$3.53	\$1,553.20
Walkways Fencing	300	Fencing, chain link, including posts and fabric - 8 ft. to 10 ft. high	300.00	LF	\$3.53	\$1,059.00
Walkways Grating	3000 sf	USER PROVIDED ITEM	3,000.00	sf	\$1.06	\$3,180.00
L-Burner Bld Stairs	900 CF	USER PROVIDED ITEM	900.00	CF	\$1.06	\$954.00
N-Secondary Crusher Stairs	1458 CF	USER PROVIDED ITEM	1,458.00	CF	\$1.06	\$1,545.48
N-Secondary Crusher Conveyors	500 cy	Conveyor, demolition, off-site disposal in approved landfill, 15 mile haul	13,500.00	CF	\$0.85	\$11,475.00
O-Homogenizing Silos Stairs	1350 CF	USER PROVIDED ITEM	1,350.00	CF	\$1.06	\$1,431.00
O-Homogenizing Silos Grating	5530 SF	USER PROVIDED ITEM	5,530.00	SF	\$1.06	\$5,861.80
P-Coal Facilities Stairs	300 CF	USER PROVIDED ITEM	300.00	CF	\$1.06	\$318.00
P-Coal Facilities Conveyors	111 CY	Conveyor, demolition, off-site disposal in approved landfill, 15 mile haul	2,997.00	CF	\$0.85	\$2,547.45
R-Bag House Hoppers	13x10x2	USER PROVIDED ITEM	2,041.00	CF	\$1.24	\$2,530.84
R-Bag House V Hopper	1x70x6	USER PROVIDED ITEM	7,560.00	CF	\$1.24	\$9,374.40
S-Raw Mat Dryer Stairs	180 CF	USER PROVIDED ITEM	180.00	CF	\$1.06	\$190.80
Y-Fuel Tank Diesel AST	10000 gal	Haul tank to certified salvage dump - 9,000 to 12,000 gal. tank	1.00	EA	\$1,050.00	\$1,050.00
Y-Fuel Tank Gas AST	5000 gal	Haul tank to certified salvage dump - 3,000 to 5,000 gal. tank	1.00	EA	\$760.00	\$760.00
Z-Kiln Conveyors	888 cy	Conveyor, demolition, off-site disposal in approved landfill, 15 mile haul	23,976.00	CF	\$0.85	\$20,379.60
EE-SNCR Fencing	192 LF	Fencing, chain link, including posts and fabric - 8 ft. to 10 ft. high	192.00	LF	\$3.53	\$677.76
EE-SNCR Tank	35000 gal	Haul tank to certified salvage dump - 9,000 to 12,000 gal. tank	1.00	EA	\$1,050.00	\$1,050.00
EE-SNCR Piping	370 lf	Pipe, steel, welded connections - 4 in. diameter pipe	370.00	LF	\$2.03	\$751.10
FF-Lime/Carbon Hoppers	7972 cf	USER PROVIDED ITEM	7,972.00	CF	\$1.24	\$9,885.28

Job Hours:	615.00	Subtotal		Total Cost	
		(unadjusted):	\$849,419.86	(adjusted for	
				location):	\$779,767.43

DEMOLITION WORK

Task description:

Table 5 Other Reclamation Costs

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 003
Date: 2/14/2024
User: JPL

State: Colorado
County: Boulder

Abbreviation: None
Filename: M208-003

Agency or organization name: DRMS

UNIT COSTS

Location adjustment: 91.80 %

Structure or Item Description	Dimensions	Demolition Menu Selection	Quantity	Unit	Unit Cost	Total Cost
Asbestos Abatement	11292	Remove cementitious material - irregular surface	11,292.00	SF	\$4.59	\$51,830.28
Asbestos Abatement - Bag	11292x0.042	Collect and bag bulk material, 3 CF bags - by hand	157.00	EA	\$12.10	\$1,899.70
Asbestos Abatement - Disposal	157 bags	Disposal charge, not including haul - Maximum	470.00	CY	\$355.00	\$166,850.00
Utility Poles	90 ea	Utility Poles, Wood 35' - 45' high (each pole)	90.00	EA	\$325.00	\$29,250.00
Utility Lines	2640	Disposal of utility pole cross arms and hardware surplus material	2,640.00	LF	\$0.01	\$26.40
Rail Removal	8142 LF	Railroad track - Ties and track	8,142.00	LF	\$16.15	\$131,493.30
Hazardous Waste Freeline Solution	30 gal	Hazardous waste removal - Drum solids/liquids, per drum, (1-6 drum job)	1.00	DRUM	\$492.56	\$492.56
Hazardous Waste Aqueous Ammonia (19%)	31,500 gal	Hazardous waste removal - Bulk liquids, large quantities (over 2,500 gal.)	31,500.00	GAL	\$1.98	\$62,404.65
Used Grease	330 gal	Hazardous waste removal - Drum solids/liquids, per drum, (1-6 drum job)	6.00	DRUM	\$492.56	\$2,955.38
Hydrochloric Acid	3 L	Hazardous waste removal - Drum solids/liquids, per drum, (1-6 drum job)	0.01	DRUM	\$492.56	\$4.93
Ethyl Alcohol	30 L	Hazardous waste removal - Drum solids/liquids, per drum, (1-6 drum job)	0.15	DRUM	\$492.56	\$73.88
Ammonium Acetate	500 g	Hazardous waste removal - Bulk solids, small quantities (up to 1.5 tons)	0.01	TON	\$1,671.09	\$16.71
Phenolphthalein	100 g	Hazardous waste removal - Bulk solids,	0.01	TON	\$1,671.09	\$16.71

		small quantities (up to 1.5 tons)				
Sodium Hydroxide	1 kg	Hazardous waste removal - Bulk solids, small quantities (up to 1.5 tons)	0.01	TON	\$1,671.09	\$16.71
Strontium Nitrate	1 kg	Hazardous waste removal - Bulk solids, small quantities (up to 1.5 tons)	0.01	TON	\$1,671.09	\$16.71
Calcium Hydroxide	500 g	Hazardous waste removal - Bulk solids, small quantities (up to 1.5 tons)	0.01	TON	\$1,671.09	\$16.71

Job Hours: 2,700.00

**Subtotal
(unadjusted):** \$447,364.63

**Total Cost
(adjusted for
location):** \$410,680.73

SCRAPER TEAM WORKTask description: **Excavation of On-Site Disposal Cell**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 004State: ColoradoAbbreviation: NoneDate: 2/14/2024County: BoulderFilename: M208-004User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	Cat D8T - 8SU
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	10	NA	NA	50	20
Ownership cost/hour:	\$281.32	\$173.32	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	\$10.97	NA	NA	\$59.82	\$23.69
%Utilization-ripper:	NA	NA	NA	NA	0	NA
Ripper own. cost/hour:	NA	\$0.00	NA	NA	\$5.75	\$0.00
Ripper op. cost/hour:	NA	\$0.00	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	\$40.04	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	\$224.33	NA	NA	\$301.66	\$135.36
Number of Units:	6	1	0	0	1	1
Group Subtotals:	Work:	\$4,173.47	Support:	\$0.00	Maint:	\$437.02

Total work team cost/hour: **\$4,610.49****MATERIAL QUANTITIES**Initial volume: 247,970

CCY

Swell factor: 1.250Loose volume: **309,963**

LCY

Source of estimated volume: Operator Original Sumbittal Table 5 not Table 6Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight: 2,650 lbs/LCY
 Material description: Decomposed rock - 25% Rock, 75% Earth
 Rated Payload: 81,600 pounds
 Payload Capacity: 30.79 LCY

Struck Volume: 24.00 LCY
 Heaped Volume: 34.00 LCY
 Average Volume: 29.00 LCY
 Adjusted Capacity: **29.00** LCY

Cycle Time:Scraper Loading Time: 1.00 MinutesManeuver and Spread Time: 0.60 MinutesJob Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Hard, smooth, stabilized, surfaced, watered, maintained 2.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	500.00	1.00	2.00	3.00	2800	0.46

Haul Time: 0.46 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	500.00	-2.00	2.00	0.00	2965	0.27

Return Time: 0.27 minutesTotal Scraper team cycle time: 2.33 minutesAdjusted for job conditions: 1,239.66 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 3,718.97 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 3,718.97 LCY/HourUnadjusted unit production/hour: 1,493.56 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 83.35 HoursUnit cost: \$1.240 /LCYTotal job cost: \$384,267

SCRAPER TEAM WORKTask description: **3 Ft Clay Liner Protective Cover**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 006State: ColoradoAbbreviation: NoneDate: 2/14/2024County: BoulderFilename: M208-006User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	NA	NA	50	20
Ownership cost/hour:	\$281.32	NA	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	NA	NA	\$59.82	\$23.69
%Utilization-ripper:	NA	NA	NA	NA	0	NA
Ripper own. cost/hour:	NA	NA	NA	NA	\$5.75	\$0.00
Ripper op. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	NA	NA	\$301.66	\$135.36
Number of Units:	6	0	0	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$0.00	Maint:	\$437.02

Total work team cost/hour: **\$4,386.16****MATERIAL QUANTITIES**Initial volume: 34,426

CCY

Swell factor: 1.220Loose volume: **42,000**

LCY

Source of estimated volume: Operator Second Submittal Table 6Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight:	<u>2,800 lbs/LCY</u>	Struck Volume:	<u>24.00</u>	LCY
Material description:	<u>Clay - Natural bed</u>	Heaped Volume:	<u>34.00</u>	LCY
Rated Payload:	<u>81,600 pounds</u>	Average Volume:	<u>29.00</u>	LCY
Payload Capacity:	<u>29.14 LCY</u>	Adjusted Capacity:	<u>29.00</u>	LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Hard, smooth, stabilized, surfaced, watered, maintained 2.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	1.00	2.00	3.00	2800	1.72

Haul Time: 1.72 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	-1.00	2.00	1.00	2963	1.47

Return Time: 1.47 minutesTotal Scraper team cycle time: 4.79 minutesAdjusted for job conditions: 603.01 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 1,809.02 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 1,809.02 LCY/HourUnadjusted unit production/hour: 726.51 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 23.22 HoursUnit cost: \$2.425 /LCYTotal job cost: \$101,833

SCRAPER TEAM WORKTask description: **Interstitial Fill**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 007State: ColoradoAbbreviation: NoneDate: 2/14/2024County: BoulderFilename: M208-007User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	Cat D8T - 8SU
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	10	NA	NA	50	20
Ownership cost/hour:	\$281.32	\$173.32	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	\$10.97	NA	NA	\$59.82	\$23.69
%Utilization-ripper:	NA	NA	NA	NA	0	NA
Ripper own. cost/hour:	NA	\$0.00	NA	NA	\$5.75	\$0.00
Ripper op. cost/hour:	NA	\$0.00	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	\$40.04	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	\$224.33	NA	NA	\$301.66	\$135.36
Number of Units:	6	1	0	0	1	1
Group Subtotals:	Work:	\$4,173.47	Support:	\$0.00	Maint:	\$437.02

Total work team cost/hour: **\$4,610.49****MATERIAL QUANTITIES**Initial volume: 21,600

CCY

Swell factor: 1.250Loose volume: **27,000**

LCY

Source of estimated volume: Operator Second Submittal Table 6Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight: 2,650 lbs/LCY
 Material description: Decomposed rock - 25% Rock, 75% Earth
 Rated Payload: 81,600 pounds
 Payload Capacity: 30.79 LCY

Struck Volume: 24.00 LCY
 Heaped Volume: 34.00 LCY
 Average Volume: 29.00 LCY
 Adjusted Capacity: **29.00** LCY

Cycle Time:Scraper Loading Time: 1.00 MinutesManeuver and Spread Time: 0.60 MinutesJob Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Hard, smooth, stabilized, surfaced, watered, maintained 2.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	1.00	2.00	3.00	2800	1.71

Haul Time: 1.71 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	-1.00	2.00	1.00	2963	1.47

Return Time: 1.47 minutesTotal Scraper team cycle time: 4.78 minutesAdjusted for job conditions: 604.27 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 1,812.80 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 1,812.80 LCY/HourUnadjusted unit production/hour: 728.03 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 14.89 HoursUnit cost: \$2.543 /LCYTotal job cost: \$68,669

SCRAPER TEAM WORKTask description: **30-inch Cover**Site: **Lyons Quarry**Permit Action: **2024 Cost Est**Permit/Job#: **M1977208****PROJECT IDENTIFICATION**Task #: **008**State: **Colorado**Abbreviation: **None**Date: **2/14/2024**County: **Boulder**Filename: **M208-008**User: **JPL**Agency or organization name: **DRMS****HOURLY EQUIPMENT**COSTShift basis: **1 per day**

Equipment Description	
-Scraper:	Cat 637G w/push-pull
-Dozer:	Cat D8T - 8SU
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	10	NA	NA	50	20
Ownership cost/hour:	\$281.32	\$173.32	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	\$10.97	NA	NA	\$59.82	\$23.69
%Utilization-ripper:	NA	NA	NA	NA	0	NA
Ripper own. cost/hour:	NA	\$0.00	NA	NA	\$5.75	\$0.00
Ripper op. cost/hour:	NA	\$0.00	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	\$40.04	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	\$224.33	NA	NA	\$301.66	\$135.36
Number of Units:	6	1	0	0	1	1
Group Subtotals:	Work:	\$4,173.47	Support:	\$0.00	Maint:	\$437.02

Total work team cost/hour: **\$4,610.49****MATERIAL QUANTITIES**Initial volume: **32,000**

CCY

Swell factor: **1.250**Loose volume: **40,000**

LCY

Source of estimated volume: **Operator Second Submittal Table 6**Source of estimated swell factor: **Cat Handbook****HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight: **2,650 lbs/LCY**
 Material description: **Decomposed rock - 25% Rock, 75% Earth**
 Rated Payload: **81,600 pounds**
 Payload Capacity: **30.79 LCY**

Struck Volume: **24.00** LCY
 Heaped Volume: **34.00** LCY
 Average Volume: **29.00** LCY
 Adjusted Capacity: **29.00** LCY

Cycle Time:Scraper Loading Time: 1.00 MinutesManeuver and Spread Time: 0.60 MinutesJob Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Hard, smooth, stabilized, surfaced, watered, maintained 2.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	1.00	2.00	3.00	2800	1.71

Haul Time: 1.71 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4000.00	-1.00	2.00	1.00	2963	1.47

Return Time: 1.47 minutesTotal Scraper team cycle time: 4.78 minutesAdjusted for job conditions: 604.27 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 1,812.80 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 1,812.80 LCY/HourUnadjusted unit production/hour: 728.03 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 22.07 HoursUnit cost: \$2.543 /LCYTotal job cost: \$101,732

Post-Mining Drainage Channel Construction (Ditches)

Task description: Stormwater Diversion Ditch

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 009

State: Colorado

Abbreviation: None

Date: 10/11/2024

County: Boulder

Filename: M208-009

User: JPL

Agency or organization name: DRMS

Channel	Length (ft)	Depth (ft)	Width (bottom) (ft)	Side Slopes (XH:1V)	Width (top) (ft)	Excavated Vol./LF (CY)	Excavated Vol. (total) (CY)
Stormwater Ditch	4,000	3.00	2.00	4.00	26.00	1.5556	6,222
RipRap Pad	10	10.00	1.50	4.00	81.50	15.3704	154
Totals:	4,010						6,376

Riprap	Riprap Thickness (2xD50) (ft)	Perimeter, P (ft)	Area for Geotextile (excl. anchor trenches) (sf)	Riprap Vol. (CY)
	0.00	26.74	106,955	0
	1.50	83.96	840	47
Totals:			107,794	47

Materials Needed:

Geotextile (SY incl. 15%
wastage): 13,774
Riprap (CY): 47
Excavation (CY): 6,376

Costs:

Material Costs:	Geotextile (SY):	\$3.30	Riprap (CY):	\$34.50	Excavation (CY):	\$0.00
Labor Cost:		\$0.32		\$14.65		\$2.84
Equipment Cost:		\$0.00		\$15.10		\$2.39
Means Reference	31 32 1916 1510		31 37 1310 0100		31 23 1642 0310	

Totals:

Geotextile (SY): \$49,860.79
Riprap (CY): \$2,996.98
Excavation (CY): \$33,346.09

Hours:

Geotextile (SY): 157.41

87.50 SY/HR	
Riprap (CY):	6.02
7.75 CY/HR	
Excavation (CY):	159.40
40.00 CY/HR	

Total Post-Mining Channel Reconstruction hours:	<u>322.83</u>	Hours
Total Post-Mining Channel Reconstruction Cost:	<u>\$86,204</u>	

BULLDOZER WORKTask description: **B-Pit Topsoil Stripping**Site: **Lyons Quarry**Permit Action: **2024 Cost Est**Permit/Job#: **M1977208****PROJECT IDENTIFICATION**Task #: **010**State: **Colorado**Abbreviation: **None**Date: **10/22/2024**County: **Boulder**Filename: **M208-010**User: **JPL**Agency or organization name: **DRMS****HOURLY EQUIPMENT COST**Basic Machine: **Cat D9T - 9SU**Horsepower: **405**Blade Type: **Semi-Universal**Attachment: **3-shank ripper**Shift Basis: **1 per day**Data Source: **(CRG)****Cost Breakdown:**

		<u>Utilization %</u>
Ownership Cost/Hour:	\$253.16	NA
Operating Cost/Hour:	\$164.35	100
Ripper own. Cost/Hour:	\$18.79	NA
Ripper op. Cost/Hour:	\$0.95	10
Operator Cost/Hour:	\$40.04	NA
Total unit Cost/Hour:	\$477.28	
Total Fleet Cost/Hour:	\$477.28	

MATERIAL QUANTITIESInitial Volume: **15,811**Swell factor: **1.429**Loose volume: **22,588 LCY**Source of estimated volume: **Operator Provided Table 7**Source of estimated swell
factor: **Cat Handbook****HOURLY PRODUCTION**Average push distance: **150 feet**Unadjusted hourly
production: **910.5 LCY/hr**Materials consistency description: **Compacted fill or embankment 0.9**Average push
gradient: **0 %**Average site altitude: **5,200 feet**Material weight: **1,600 lbs/LCY**Weight description: **Top Soil****Job Condition Correction Factor**Operator Skill: **0.750****Source****(AVG.)**

Material consistency:	0.900	(CAT HB))
Dozing method:	1.200	(SLOT)
Visibility:	1.000	(AVG.)
Job efficiency:	0.830	(1 SHIFT/DAY)
Spoil pile:	1.000	(DOZ-OC)
Push gradient:	1.000	(CAT HB)
Altitude:	1.000	(CAT HB)
Material Weight:	1.438	(CAT HB)
Blade type:	1.000	(PAT)

Net correction: 0.9668

Adjusted unit
production: 880.27 LCY/hr

Adjusted fleet
production: **880.27** LCY/hr

JOB TIME AND COST

Fleet size: 1 Dozer(s)

Unit cost: \$0.542/LCY

Total job time: **25.66** Hours

Total job cost: **\$12,247**

TRUCK/LOADER TEAM WORKTask description: **Backfill C Pit**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 011State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-011User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT COST**Shift basis: 1 per day

	Equipment Description
Truck Loader Team -Truck:	Cat 777F
-Loader:	CAT 992K
Support Equipment -Load Area:	NA
-Dump Area:	Cat D9T - 9SU
Road Maintenance –Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Truck/Loader Team		Support Equipment		Maintenance Equipment	
	Truck	Loader	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	100	NA	50	30	30
Ownership cost/hour:	\$199.47	\$270.75	NA	\$253.16	\$179.39	\$111.67
Operating cost/hour:	\$152.44	\$230.31	NA	\$82.18	\$35.89	\$35.53
%Utilization-riper:	NA	0	NA	NA	NA	NA
Ripper own. cost/hour:	NA	\$0.00	NA	\$0.00	\$0.00	\$0.00
Ripper op. cost/hour:	NA	\$0.00	NA	\$0.00	\$0.00	\$0.00
Operator cost/hour:	\$24.82	\$56.84	NA	\$40.04	\$56.70	\$0.00
Unit Subtotals:	\$376.73	\$557.90	NA	\$375.37	\$271.98	\$147.20
Number of Units:	4	2	0	1	1	1
Group Subtotals:	Work: \$2,622.72		Support: \$375.37		Maint: \$419.18	

Total work team cost/hour: **\$3,417.27****MATERIAL QUANTITIES**Initial volume: 205,158

CCY

Swell factor: 1.000Loose volume: **205,158**

LCY

Source of estimated volume: Operator submitted Table 7Source of estimated swell factor: Cat HandbookMaterial Purchase Cost: \$0.00Total Cost: \$0.00**HOURLY PRODUCTION****Truck Capacity:**

Truck Payload (weight) Basis:

Material weight: 2,650 Pounds/LCYDescription: Decomposed rock - 25% Rock, 75% EarthRated Payload: 200,000 Pounds

Payload Capacity: 75.47 LCY

Truck Bed (volume) Basis:

Struck Volume: 60.60 LCY
 Heaped Volume: 78.80 LCY
 Average Volume: 69.70 LCY
 Adjusted Volume: 75.47 LCY

Final Truck Volume Based on Number of Loader Passes: 65.60 LCY

Loading Tool Capacity

Bucket Size Class: NA

Rated Capacity:	<u>16.000</u>	LCY (heaped)
Bucket Fill Factor:	<u>1.025</u>	Rock - Earth Mixture (100%-105%) 1.025
Adjusted Capacity:	<u>16.400</u>	LCY

Job Condition Corrections:

Site Altitude (ft.): 5200 feet

	Truck	Loader	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Loading Tool Cycle Time:

Number of Loading Tool Passes Required to Fill
 Truck: 4 passes

Excavators and Front Shovels:

Machine Cycle Time vs. Job Condition Rating: NA
 Selected Value within this Basic Rating: NA

Track Loaders – Material Description: _____

Cycle Time Elements (min.):

Load: NA Maneuver: NA Dump: 0.100

Wheel and Track Loaders - Unadjusted Basic Loader Cycle Time (load, dump, maneuver): 0.625 minutes

Cycle Time Factors		Factor (min.)	Source
Material:	Material 3/4" to 6" diameter 0.00	0.000	(Cat HB)
Stockpile:	No adjustment - factor not applicable 0.00	0.000	(Cat HB)
Truck Ownership:	No adjustment - factor not applicable 0.00	0.000	(Cat HB)
Operation:	Inconsistent operation 0.04	0.040	(Cat HB)
Dump Target:	Nominal target 0.00	0.000	(Cat HB)
Net Cycle Time Adjustment:		0.040	minutes
Adjusted Loader Cycle Time:		0.665	minutes
Net Load Time per Truck:		2.095	minutes

Truck Cycle Time:

Truck Exchange Time:	<u>0.80</u>	Minutes	Adjusted for site altitude:	<u>0.800</u>	Minutes
Truck Load Time:	<u>2.095</u>	Minutes	Adjusted for site altitude:	<u>2.095</u>	Minutes
Truck Maneuver and Dump Time:	<u>1.20</u>	Minutes	Adjusted for site altitude:	<u>1.200</u>	Minutes

Truck Travel (Haul & Return) Time: maintained 3.0

Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered,

Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	3000.00	1.00	3.00	4.00	1786	1.916

Haul Time: **1.916** minutes

Return Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	3000.00	-1.00	3.00	2.00	3503	1.082

Return Time: **1.082** minutesTotal Truck Cycle Time: **7.093** minutes

Loading Tool unit

Production 1,359.59 LCY/Hour Adjusted for job efficiency: 1,128.46 LCY/Hour
Truck Unit Production 554.91 LCY/Hour Adjusted for job efficiency: 460.58 LCY/Hour

Optimal No. of Trucks: 2 Truck(s) Selected Number of Trucks: 2 Truck(s)

Adjusted hourly truck team production: 921.16 LCY/Hour
Adjusted single truck/loader team production: 921.16 LCY/Hour
Adjusted multiple truck/loader team production: **1,842.31** LCY/Hour

JOB TIME AND COST

Fleet size: 2 Team(s) Total job time: **111.36** Hours

Unit cost: \$1.855 /LCY Total job cost: **\$380,544**

SCRAPER TEAM WORKTask description: Shale from SE Corner of site to C-PitSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 012State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-012User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	Cat D9T - 9SU
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	25	NA	50	25
Ownership cost/hour:	\$281.32	NA	\$253.16	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	\$41.09	NA	\$59.82	\$29.61
%Utilization-ripper:	NA	NA	NA	NA	NA	NA
Ripper own. cost/hour:	NA	NA	\$0.00	NA	\$0.00	\$0.00
Ripper op. cost/hour:	NA	NA	\$0.00	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	\$40.04	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	\$334.28	NA	\$295.91	\$141.28
Number of Units:	6	0	1	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$334.28	Maint:	\$437.19

Total work team cost/hour: \$4,720.61**MATERIAL QUANTITIES**Initial volume: 36,067

CCY

Swell factor: 1.333Loose volume: 48,088

LCY

Source of estimated volume: Operator Provided Table 7Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION**Scraper Bowl (volume) Basis:

Material weight: 2,100 lbs/LCY
 Material description: Shale
 Rated Payload: 81,600 pounds
 Payload Capacity: 38.86 LCY

Struck Volume: 24.00 LCY
 Heaped Volume: 34.00 LCY
 Average Volume: 29.00 LCY
 Adjusted Capacity: 29.00 LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered, maintained 3.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4500.00	2.00	3.00	5.00	1867	2.53

Haul Time: 2.53 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	4500.00	-2.00	3.00	1.00	2963	1.64

Return Time: 1.64 minutes

Total Scraper team cycle time: 5.77 minutes
 Adjusted for job conditions: 500.59 LCY/Hour
 Selected Number of Scrapers: 6 Scraper(s)
 Adjusted single scraper team (unit) hourly production: 1,501.77 LCY/Hour
 Adjusted multiple scraper team (fleet) hourly production: 1,501.77 LCY/Hour

Unadjusted unit production/hour: 603.12 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s) Total job time: 32.02 HoursUnit cost: \$3.143 /LCY Total job cost: \$151,159

MOTOR GRADER WORK

Task description: Rough Grade Plant Area

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 013

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-013

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$56.70	NA
Total Unit Cost/Hour:	\$361.48	
Total Fleet Cost/Hour:	\$361.48	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 31.10 acres

Source of estimated acreage: Operator Provided Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>3.25</u>	mph
Selected Application:	<u>Heavy blading (0-6 mph) - 3.25</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>5.5152</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.90</u>	(1sh/d, fav.)
Net Correction:	<u>0.9000</u>	multiplier

Adjusted Hourly Unit Production: 4.9636 acres/Hour

Adjusted Hourly Fleet Production: 4.9636 acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 6.27 Hours

Unit cost: \$72.83 per acre Total job cost: \$2,265

MOTOR GRADER WORK

Task description: Fine Grade Plant Area

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 014

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-014

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$57.29	NA
Total Unit Cost/Hour:	\$362.07	
Total Fleet Cost/Hour:	\$362.07	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 31.10 acres

Source of estimated acreage: Operator Provided Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>1.50</u>	mph
Selected Application:	<u>Finish grading (0-2.5 mph) - 1.5</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>2.5455</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.90</u>	(1sh/d, fav.)
Net Correction:	<u>0.9000</u>	multiplier

Adjusted Hourly Unit Production: 2.2909 acres/Hour

Adjusted Hourly Fleet Production: 2.2909 acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 13.58 Hours

Unit cost: \$158.05 per acre Total job cost: \$4,915

MOTOR GRADER WORK

Task description: Rough Grade C-Pit

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 015

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-015

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$56.70	NA
Total Unit Cost/Hour:	\$361.48	
Total Fleet Cost/Hour:	\$361.48	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 27.00 acres

Source of estimated acreage: Operator Provided Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>3.25</u>	mph
Selected Application:	<u>Heavy blading (0-6 mph) - 3.25</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>5.5152</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.85</u>	(1sh/d, mod.)
Net Correction:	<u>0.8500</u>	multiplier

Adjusted Hourly Unit Production: 4.6879 acres/Hour

Adjusted Hourly Fleet Production: 4.6879 acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 5.76 Hours

Unit cost: \$77.11 per acre Total job cost: \$2,082

MOTOR GRADER WORK

Task description: Fine Grade C-Pit

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 016

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-016

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$57.29	NA
Total Unit Cost/Hour:	\$362.07	
Total Fleet Cost/Hour:	\$362.07	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 27.00 acres

Source of estimated acreage: Operator Provided Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>1.50</u>	mph
Selected Application:	<u>Finish grading (0-2.5 mph) - 1.5</u>	
Selected Blade Angle:	<u>-1</u>	degrees
Effective Blade Length:	<u>0.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>9.75</u>	feet
Unadjusted Hourly Unit Production:	<u>1.7727</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.85</u>	(1sh/d, mod.)
Net Correction:	<u>0.8500</u>	multiplier

Adjusted Hourly Unit Production: 1.5068 acres/Hour

Adjusted Hourly Fleet Production: **1.5068** acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: **17.92** Hours

Unit cost: \$240.29 per acre Total job cost: **\$6,488**

BULLDOZER RIPPING WORK

Task description: Rip Roads and Facilities

Site: Lyons Quarry Permit Action: 2024 Cost Est Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 017 State: Colorado Abbreviation: None
Date: 10/22/2024 County: Boulder Filename: M208-017
User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: Cat D9T - 9SU Horsepower: 405
Ripper Attachment: 3-Shank Ripper Shift Basis: 1 per day
Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$253.16	NA
Operating Cost/Hour:	\$164.35	100
Ripper Ownership Cost/Hour:	\$18.79	NA
Ripper Operating Cost/Hour:	\$9.48	100
Operator Cost/Hour:	\$40.04	NA
Total Unit Cost/Hour:	\$485.82	
Total Fleet Cost/Hour:	\$485.82	

MATERIAL QUANTITIES

Selected estimating method: Area

Alternate Methods:

Seismic: NA Bank Volume: NA BCY NA
Area: 4.00 acres Rip Depth (ft): 1.00 Volume: 6,453 BCY or CCY

Source of estimated quantity: Operator Provided Table 8

HOURLY PRODUCTION

Seismic:

Seismic Velocity: NA feet/second

Area:

Average Ripping Depth: 2.63 feet/pass
Average Ripping Width: 7.67 feet/pass
Average Ripping Length: 500.00 feet/pass
Average Dozer Speed: 88.00 feet/minute
Average Maneuver Time: 0.25 minutes/pass
Production per unit area: 0.891 acres/hour

Job Condition Correction Factors

Unadjusted Hourly Unit Production: 0.891 Acres/hr
Site Altitude: 5,200 feet
Altitude Adj: 1.00 (CAT HB)
Job Efficiency: 0.83 (1 shift/day)
Net Correction: 0.83 multiplier

Adjusted Hourly Unit Production: 0.74 Acres/hr
Adjusted Hourly Fleet Production: 0.74 Acres/hr

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 5.41 Hours

Unit cost: \$657.289 Per acre Total job cost: \$2,629

MOTOR GRADER WORK

Task description: Grade Roads and Facilities

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 018

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-018

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$57.29	NA
Total Unit Cost/Hour:	\$362.07	
Total Fleet Cost/Hour:	\$362.07	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 4.00 acres

Source of estimated acreage: Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>1.50</u>	mph
Selected Application:	<u>Finish grading (0-2.5 mph) - 1.5</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>2.5455</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.90</u>	(1sh/d, fav.)
Net Correction:	<u>0.9000</u>	multiplier

Adjusted Hourly Unit Production: 2.2909 acres/Hour

Adjusted Hourly Fleet Production: 2.2909 acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 1.75 Hours

Unit cost: \$158.05 per acre Total job cost: \$632

BULLDOZER RIPPING WORK

Task description: Rip Disposal Cell

Site: Lyons Quarry Permit Action: 2024 Cost Est Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 019 State: Colorado Abbreviation: None
Date: 10/22/2024 County: Boulder Filename: M208-019
User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: Cat D9T - 9SU Horsepower: 405
Ripper Attachment: 3-Shank Ripper Shift Basis: 1 per day
Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$253.16	NA
Operating Cost/Hour:	\$164.35	100
Ripper Ownership Cost/Hour:	\$18.79	NA
Ripper Operating Cost/Hour:	\$9.48	100
Operator Cost/Hour:	\$40.04	NA
Total Unit Cost/Hour:	\$485.82	
Total Fleet Cost/Hour:	\$971.63	

MATERIAL QUANTITIES

Selected estimating method: Area

Alternate Methods:

Seismic: NA Bank Volume: NA BCY NA
Area: 13.30 acres Rip Depth (ft): 0.50 Volume: 10,729 BCY or CCY

Source of estimated quantity: Operator Provided Table 8

HOURLY PRODUCTION

Seismic:

Seismic Velocity: NA feet/second

Area:

Average Ripping Depth: 1.00 feet/pass
Average Ripping Width: 7.67 feet/pass
Average Ripping Length: 500.00 feet/pass
Average Dozer Speed: 88.00 feet/minute
Average Maneuver Time: 0.25 minutes/pass
Production per unit area: 0.891 acres/hour

Job Condition Correction Factors

Unadjusted Hourly Unit Production: 0.891 Acres/hr
Site Altitude: 5,200 feet
Altitude Adj: 1.00 (CAT HB)
Job Efficiency: 0.83 (1 shift/day)
Net Correction: 0.83 multiplier

Adjusted Hourly Unit Production: 0.74 Acres/hr
Adjusted Hourly Fleet Production: 1.48 Acres/hr

JOB TIME AND COST

Fleet size: 2 Grader(s) Total job time: 9.00 Hours

Unit cost: \$657.289 Per acre Total job cost: \$8,742

MOTOR GRADER WORK

Task description: Grade Disposal Cell

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 020

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-020

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$57.29	NA
Total Unit Cost/Hour:	\$362.07	
Total Fleet Cost/Hour:	\$362.07	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 13.30 acres

Source of estimated acreage: Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>1.50</u>	mph
Selected Application:	<u>Finish grading (0-2.5 mph) - 1.5</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>2.5455</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.85</u>	(1sh/d, mod.)
Net Correction:	<u>0.8500</u>	multiplier

Adjusted Hourly Unit Production: 2.1636 acres/Hour

Adjusted Hourly Fleet Production: **2.1636** acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: **6.15** Hours

Unit cost: \$167.34 per acre Total job cost: **\$2,226**

MOTOR GRADER WORK

Task description: Grade B-Pit Borrow

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 021 State: Colorado Abbreviation: None
Date: 10/22/2024 County: Boulder Filename: M208-021
User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M Horsepower: 297
Ripper Attachment: Multi-Shank Ripper Shift Basis: 1 per day
Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	<u>\$179.39</u>	<u>NA</u>
Operating Cost/Hour:	<u>\$119.64</u>	<u>100</u>
Ripper Ownership Cost/Hour:	<u>\$5.75</u>	<u>NA</u>
Ripper Operating Cost/Hour:	<u>\$0.00</u>	<u>0</u>
Operator Cost/Hour:	<u>\$57.29</u>	<u>NA</u>
Total Unit Cost/Hour:	<u>\$362.07</u>	
Total Fleet Cost/Hour:	<u>\$362.07</u>	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 21.00 acres

Source of estimated acreage: Table 8

HOURLY PRODUCTION

Average Grader Speed: 1.50 mph
Selected Application: Finish grading (0-2.5 mph) - 1.5
Selected Blade Angle: 0 degrees
Effective Blade Length: 16.00 feet
Width of blade overlap per pass: 2.00 feet
Net grading or ripping width per pass: 14.00 feet
Unadjusted Hourly Unit Production: 2.5455 acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	<u>(CAT HB)</u>
Job Efficiency:	<u>0.85</u>	<u>(1sh/d, mod.)</u>
Net Correction:	<u>0.8500</u>	<u>multiplier</u>

Adjusted Hourly Unit Production: 2.1636 acres/Hour
Adjusted Hourly Fleet Production: 2.1636 acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: 9.71 Hours

Unit cost: \$167.34 per acre Total job cost: \$3,514

MOTOR GRADER WORK

Task description: Grade Topsoil Stockpile

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 022

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-022

User: JPL

Agency or organization name: DRMS

HOURLY EQUIPMENT COST

Basic Machine: CAT 16M

Horsepower: 297

Ripper Attachment: Multi-Shank Ripper

Shift Basis: 1 per day

Data Source: (CRG)

Cost Breakdown:

		Utilization %
Ownership Cost/Hour:	\$179.39	NA
Operating Cost/Hour:	\$119.64	100
Ripper Ownership Cost/Hour:	\$5.75	NA
Ripper Operating Cost/Hour:	\$0.00	0
Operator Cost/Hour:	\$57.29	NA
Total Unit Cost/Hour:	\$362.07	
Total Fleet Cost/Hour:	\$362.07	

MATERIAL QUANTITIES

Total Area to be graded or ripped: 8.70 acres

Source of estimated acreage: Table 8

HOURLY PRODUCTION

Average Grader Speed:	<u>1.50</u>	mph
Selected Application:	<u>Finish grading (0-2.5 mph) - 1.5</u>	
Selected Blade Angle:	<u>0</u>	degrees
Effective Blade Length:	<u>16.00</u>	feet
Width of blade overlap per pass:	<u>2.00</u>	feet
Net grading or ripping width per pass:	<u>14.00</u>	feet
Unadjusted Hourly Unit Production:	<u>2.5455</u>	acres/hour

Job Condition Correction Factors

Site Altitude: 5200 feet

		Source
Altitude Adj:	<u>1.00</u>	(CAT HB)
Job Efficiency:	<u>0.85</u>	(1sh/d, mod.)
Net Correction:	<u>0.8500</u>	multiplier

Adjusted Hourly Unit Production: 2.1636 acres/Hour

Adjusted Hourly Fleet Production: **2.1636** acres/Hour

JOB TIME AND COST

Fleet size: 1 Grader(s) Total job time: **4.02** Hours

Unit cost: \$167.34 per acre Total job cost: **\$1,456**

SCRAPER TEAM WORKTask description: Topsoil Plant AreaSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 023State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-023User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	NA	NA	50	25
Ownership cost/hour:	\$281.32	NA	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	NA	NA	\$59.82	\$29.61
%Utilization-ripper:	NA	NA	NA	NA	NA	NA
Ripper own. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Ripper op. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	NA	NA	\$295.91	\$141.28
Number of Units:	6	0	0	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$0.00	Maint:	\$437.19

Total work team cost/hour: \$4,386.33**MATERIAL QUANTITIES**Initial volume: 26,344

CCY

Swell factor: 1.429Loose volume: 37,635

LCY

Source of estimated volume: Table 9Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION**Scraper Bowl (volume) Basis:

Material weight: 1,600 lbs/LCY
 Material description: Top Soil
 Rated Payload: 81,600 pounds
 Payload Capacity: 51.00 LCY

Struck Volume: 24.00 LCY
 Heaped Volume: 34.00 LCY
 Average Volume: 29.00 LCY
 Adjusted Capacity: 29.00 LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered, maintained 3.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	5000.00	2.00	3.00	5.00	1867	2.78

Haul Time: 2.78 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	5000.00	-2.00	3.00	1.00	2963	1.80

Return Time: 1.80 minutesTotal Scraper team cycle time: 6.18 minutesAdjusted for job conditions: 467.38 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 1,402.14 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 1,402.14 LCY/HourUnadjusted unit production/hour: 563.11 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 26.84 HoursUnit cost: \$3.128 /LCYTotal job cost: \$117,734

SCRAPER TEAM WORKTask description: Topsoil C-PitSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 024State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-024User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	NA	NA	50	25
Ownership cost/hour:	\$281.32	NA	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	NA	NA	\$59.82	\$29.61
%Utilization-ripper:	NA	NA	NA	NA	NA	NA
Ripper own. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Ripper op. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	NA	NA	\$295.91	\$141.28
Number of Units:	6	0	0	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$0.00	Maint:	\$437.19

Total work team cost/hour: \$4,386.33**MATERIAL QUANTITIES**Initial volume: 22,870

CCY

Swell factor: 1.429Loose volume: 32,672

LCY

Source of estimated volume: Table 9Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight:	<u>1,600 lbs/LCY</u>	Struck Volume:	<u>24.00</u>	LCY
Material description:	<u>Top Soil</u>	Heaped Volume:	<u>34.00</u>	LCY
Rated Payload:	<u>81,600 pounds</u>	Average Volume:	<u>29.00</u>	LCY
Payload Capacity:	<u>51.00 LCY</u>	Adjusted Capacity:	<u>29.00</u>	LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered, maintained 3.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	5000.00	2.00	3.00	5.00	1867	2.78

Haul Time: 2.78 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	5000.00	-2.00	3.00	1.00	2963	1.80

Return Time: 1.80 minutesTotal Scraper team cycle time: 6.18 minutesAdjusted for job conditions: 467.38 LCY/HourSelected Number of Scrapers: 6 Scraper(s)Adjusted single scraper team (unit) hourly production: 1,402.14 LCY/HourAdjusted multiple scraper team (fleet) hourly production: 1,402.14 LCY/HourUnadjusted unit production/hour: 563.11 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s)Total job time: 23.30 HoursUnit cost: \$3.128 /LCYTotal job cost: \$102,209

SCRAPER TEAM WORKTask description: **Topsoil Roads and Facilities**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 025State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-025User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

Equipment Description	
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	NA	NA	50	25
Ownership cost/hour:	\$281.32	NA	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	NA	NA	\$59.82	\$29.61
%Utilization-ripper:	NA	NA	NA	NA	NA	NA
Ripper own. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Ripper op. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	NA	NA	\$295.91	\$141.28
Number of Units:	6	0	0	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$0.00	Maint:	\$437.19

Total work team cost/hour: **\$4,386.33****MATERIAL QUANTITIES**Initial volume: 3,390

CCY

Swell factor: 1.429Loose volume: **4,843**

LCY

Source of estimated volume: Table 9Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight:	<u>1,600 lbs/LCY</u>	Struck Volume:	<u>24.00</u>	LCY
Material description:	<u>Top Soil</u>	Heaped Volume:	<u>34.00</u>	LCY
Rated Payload:	<u>81,600 pounds</u>	Average Volume:	<u>29.00</u>	LCY
Payload Capacity:	<u>51.00 LCY</u>	Adjusted Capacity:	<u>29.00</u>	LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered, maintained 3.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	2500.00	2.00	3.00	5.00	1867	1.44

Haul Time: 1.44 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	2500.00	-2.00	3.00	1.00	2963	0.96

Return Time: 0.96 minutes

Total Scraper team cycle time: 4.00 minutes
 Adjusted for job conditions: 722.10 LCY/Hour
 Selected Number of Scrapers: 6 Scraper(s)
 Adjusted single scraper team (unit) hourly production: 2,166.30 LCY/Hour
 Adjusted multiple scraper team (fleet) hourly production: 2,166.30 LCY/Hour

Unadjusted unit production/hour: 870.00 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s) Total job time: 2.24 HoursUnit cost: \$2.025 /LCY Total job cost: \$9,806

SCRAPER TEAM WORKTask description: Topsoil Disposal CellSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 026State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-026User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT**COSTShift basis: 1 per day

	Equipment Description
-Scraper:	Cat 637G w/push-pull
-Dozer:	NA
Support Equipment -Load Area:	NA
-Dump Area:	NA
Road Maintenance -Motor Grader:	CAT 16M
-Water Truck:	Water Tanker, 10,000 Gal.

Cost Breakdown:

	Scraper Work Team		Support Equipment		Maintenance Equipment	
	Scraper	Dozer	Load Area	Dump Area	Motor Grader	Water Truck
%Utilization-machine:	100	NA	NA	NA	50	25
Ownership cost/hour:	\$281.32	NA	NA	NA	\$179.39	\$111.67
Operating cost/hour:	\$319.35	NA	NA	NA	\$59.82	\$29.61
%Utilization-ripper:	NA	NA	NA	NA	NA	NA
Ripper own. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Ripper op. cost/hour:	NA	NA	NA	NA	\$0.00	\$0.00
Operator cost/hour:	\$57.52	NA	NA	NA	\$56.70	\$0.00
Unit Subtotals:	\$658.19	NA	NA	NA	\$295.91	\$141.28
Number of Units:	6	0	0	0	1	1
Group Subtotals:	Work:	\$3,949.14	Support:	\$0.00	Maint:	\$437.19

Total work team cost/hour: \$4,386.33**MATERIAL QUANTITIES**Initial volume: 11,270

CCY

Swell factor: 1.429Loose volume: 16,100

LCY

Source of estimated volume: Table 9Source of estimated swell factor: Cat Handbook**HOURLY PRODUCTION****Scraper Bowl (volume) Basis:**

Material weight: 1,600 lbs/LCY
 Material description: Top Soil
 Rated Payload: 81,600 pounds
 Payload Capacity: 51.00 LCY

Struck Volume: 24.00 LCY
 Heaped Volume: 34.00 LCY
 Average Volume: 29.00 LCY
 Adjusted Capacity: 29.00 LCY

Cycle Time:

Scraper Loading Time: 1.00 Minutes
 Maneuver and Spread Time: 0.60 Minutes

Job Condition Correction:

Site Altitude: 5200 feet

	Scraper	Push Dozer	Source
Altitude Adj:	1.000	1.000	(CAT HB)
Job Efficiency:	0.830	0.830	(CAT HB)
Net Correction:	0.830	0.830	

Travel Time:Road Condition: Firm, smooth, rolling, dirt/lt. surfaced, watered, maintained 3.0Haul Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	2500.00	2.00	3.00	5.00	1867	1.44

Haul Time: 1.44 minutesReturn Route:

Seg #	Haul Distance (Ft)	Grade (%)	Roll. Res (%)	Total Res (%)	Velocity (fpm)	Travel Time (min)
1	2500.00	-2.00	3.00	1.00	2963	0.96

Return Time: 0.96 minutes

Total Scraper team cycle time: 4.00 minutes
 Adjusted for job conditions: 722.10 LCY/Hour
 Selected Number of Scrapers: 6 Scraper(s)
 Adjusted single scraper team (unit) hourly production: 2,166.30 LCY/Hour
 Adjusted multiple scraper team (fleet) hourly production: 2,166.30 LCY/Hour

Unadjusted unit production/hour: 870.00 LCY/Hour

Optimal Number of Scrapers per push dozer: _____

JOB TIME AND COSTFleet size: 1 Team(s) Total job time: 7.43 HoursUnit cost: \$2.025 /LCY Total job cost: \$32,600

BULLDOZER WORKTask description: Topsoil B-Pit BorrowSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 027State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-027User: JPLAgency or organization name: DRMS**HOURLY EQUIPMENT COST**Basic Machine: Cat D9T - 9SUHorsepower: 405Blade Type: Semi-UniversalAttachment: 3-shank ripperShift Basis: 1 per dayData Source: (CRG)**Cost Breakdown:**

		<u>Utilization %</u>
Ownership Cost/Hour:	\$253.16	NA
Operating Cost/Hour:	\$164.35	100
Ripper own.	\$18.79	NA
Cost/Hour:		
Ripper op. Cost/Hour:	\$0.47	5
Operator Cost/Hour:	\$40.04	NA

Total unit Cost/Hour: \$476.81Total Fleet Cost/Hour: \$953.62**MATERIAL QUANTITIES**Initial Volume: 22,587Swell factor: 1.000Loose volume: 22,587 LCYSource of estimated volume: Division of Reclamation, Mining & Safety

Source of estimated swell

factor: Cat Handbook**HOURLY PRODUCTION**Average push distance: 150 feet

Unadjusted hourly

production: 910.5 LCY/hrMaterials consistency description: Compacted fill or embankment 0.9

Average push

gradient: 0 %Average site altitude: 5,200 feetMaterial weight: 1,600 lbs/LCYWeight description: Top Soil**Job Condition Correction Factor**Operator Skill: 0.750**Source****(AVG.)**

Material consistency:	0.900	(CAT HB))
Dozing method:	1.000	(GEN.)
Visibility:	1.000	(AVG.)
Job efficiency:	0.830	(1 SHIFT/DAY)
Spoil pile:	0.800	(FND-RF)
Push gradient:	1.000	(CAT HB)
Altitude:	1.000	(CAT HB)
Material Weight:	1.438	(CAT HB)
Blade type:	1.000	(PAT)

Net correction: 0.6445

Adjusted unit
production: 586.82 LCY/hr

Adjusted fleet
production: **1173.64** LCY/hr

JOB TIME AND COST

Fleet size: 2 Dozer(s)

Unit cost: \$0.813/LCY

Total job time: **19.25** Hours

Total job cost: **\$18,353**

REVEGETATION WORKTask description: Plant Area RevegetationSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 028State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-028User: JPLAgency or organization name: DRMS**FERTILIZING****Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres: 31.1 Cost /Acre: \$1,791.30
 Estimated Failure Rate: 25% Cost /Acre*: \$1,717.82
 *Selected Replanting Work Items: TILLING,SEEDING,MULCHING

Initial Job Cost: \$55,709.43
 Reseeding Job Cost: \$13,356.05
 Total Job Cost: \$69,065
 Job Hours: 53.00

REVEGETATION WORKTask description: On-Site Disposal Cell RevegetationSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208PROJECT IDENTIFICATIONTask #: 029State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-029User: JPLAgency or organization name: DRMSFERTILIZING**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres: 15.4 Cost /Acre: \$1,791.30
 Estimated Failure Rate: 25% Cost /Acre*: \$1,717.82
 *Selected Replanting Work Items: TILLING,SEEDING,MULCHING

Initial Job Cost: \$27,586.02
 Reseeding Job Cost: \$6,613.61
 Total Job Cost: \$34,200
 Job Hours: 26.00

REVEGETATION WORKTask description: **C-Pit Area Revegetation**Site: **Lyons Quarry**Permit Action: **2024 Cost Est**Permit/Job#: **M1977208****PROJECT IDENTIFICATION**Task #: **030**State: **Colorado**Abbreviation: **None**Date: **10/22/2024**County: **Boulder**Filename: **M208-030**User: **JPL**Agency or organization name: **DRMS****FERTILIZING****Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres: 27 Cost /Acre: \$1,791.30
 Estimated Failure Rate: 25% Cost /Acre*: \$1,717.82
 *Selected Replanting Work Items: TILLING,SEEDING,MULCHING

Initial Job Cost: \$48,365.10
 Reseeding Job Cost: \$11,595.29
 Total Job Cost: \$59,960
 Job Hours: 46.00

REVEGETATION WORKTask description: **Roads and Facilities Revegetation**Site: **Lyons Quarry**Permit Action: 2024 Cost EstPermit/Job#: M1977208**PROJECT IDENTIFICATION**Task #: 031State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-031User: JPLAgency or organization name: DRMS**FERTILIZING****Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres: 4 Cost /Acre: \$1,791.30
 Estimated Failure Rate: 25% Cost /Acre*: \$1,717.82
 *Selected Replanting Work Items: TILLING,SEEDING,MULCHING

Initial Job Cost: \$7,165.20
 Reseeding Job Cost: \$1,717.82
 Total Job Cost: \$8,883
 Job Hours: 7.00

REVEGETATION WORKTask description: B-Pit Borrow AreaSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208PROJECT IDENTIFICATIONTask #: 032State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-032User: JPLAgency or organization name: DRMSFERTILIZINGMaterials

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres: 21 Cost /Acre: \$1,791.30
 Estimated Failure Rate: 25% Cost /Acre*: \$1,717.82
 *Selected Replanting Work Items: TILLING,SEEDING,MULCHING

Initial Job Cost: \$37,617.30
 Reseeding Job Cost: \$9,018.56
 Total Job Cost: \$46,636
 Job Hours: 36.00

REVEGETATION WORKTask description: Topsoil Stockpile RevegetationSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208PROJECT IDENTIFICATIONTask #: 033State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-033User: JPLAgency or organization name: DRMSFERTILIZING**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
6-24-24, 10-20-10, 15-15-15	67.00	pound	\$0.45	\$30.36
			Total Fertilizer Materials Cost/Acre	\$30.36

Application

Description	Cost /Acre
Tractor towed spreader (MEANS 32 01 90.13 0120)	\$43.12
Total Fertilizer Application Cost/Acre	\$43.12

TILLING

Description	Cost /Acre
Disc harrowing, 6" deep (MEANS 32 91 13.23 6100)	\$117.61
Total Tilling Cost/Acre	\$117.61

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
Buffalograss - Bison	2.92	2.82	\$67.32
Blue Grama - Hachita	0.59	9.63	\$16.90
Indiangrass - Cheyenne	0.93	2.84	\$11.44
Sand Dropseed	0.03	3.58	\$0.39
Sideoats Grama - Butte	1.71	5.61	\$41.31
Sandberg Bluegrass - VNS	0.36	7.64	\$5.20
Coneflower, Prairie	0.09	2.45	\$3.20
Slender Wheatgrass - Pryor	2.05	7.48	\$12.67
Coneflower, Purple	0.31	0.59	\$11.89
Western Wheatgrass - Arriba	2.97	7.50	\$26.83
Rabbitbrush, Rubber	0.16	2.38	\$13.34

Needlegrass, Green - Lodorm	1.80	7.48	\$15.56
Flax, Lewis Blue	0.38	2.52	\$16.07
Saltbush, Four Wing	1.26	1.74	\$25.04
Yarrow, Western	0.01	0.61	\$0.48
Bluebunch Wheatgrass - Goldar	2.18	7.01	\$24.98
Totals Seed Mix	17.75	71.87	\$292.64

Application

Description	Cost /Acre
Drill Seeding (DRMS Survey Cost)	\$236.64
Total Seed Application Cost/Acre	\$236.64

MULCHING and MISCELLANEOUS

Materials

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
Hay, delivered {MEANS 31 25 14.16 1200}	2.00	TON	\$492.78	\$985.56
Total Mulch Materials Cost/Acre				\$985.56

Application

Description	Cost /Acre
Crimping, with tractor {DMG survey data}	\$85.37
Total Mulch Application Cost/Acre	\$85.37

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres:	8.7	Cost /Acre:	\$1,791.30
Estimated Failure Rate:	25%	Cost /Acre*:	\$1,717.82
*Selected Replanting Work Items:	TILLING,SEEDING,MULCHING		
Initial Job Cost:	\$15,584.31		
Reseeding Job Cost:	\$3,736.26		
Total Job Cost:	\$19,321		
Job Hours:	15.00		

REVEGETATION WORKTask description: Weed ControlSite: Lyons QuarryPermit Action: 2024 Cost EstPermit/Job#: M1977208PROJECT IDENTIFICATIONTask #: 034State: ColoradoAbbreviation: NoneDate: 10/22/2024County: BoulderFilename: M208-034User: JPLAgency or organization name: DRMSFERTILIZING**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
			\$	\$
			Total Fertilizer Materials Cost/Acre	\$0.00

Application

Description	Cost /Acre
	\$
Total Fertilizer Application Cost/Acre	\$0.00

TILLING

Description	Cost /Acre
Weed control spraying (MEANS 31 31 16.13 3100)	\$338.80
Total Tilling Cost/Acre	\$338.80

SEEDING

Seed Mix	Rate – PLS LBS / Acre	Seeds per SQ. FT	Cost /Acre
			\$
Totals Seed Mix	0.00	0.00	\$0.00

Application

Description	Cost /Acre
	\$

Total Seed Application Cost/Acre	\$0.00
---	---------------

MULCHING and MISCELLANEOUS**Materials**

Description	Units / Acre	Unit	Cost / Unit	Cost /Acre
			\$	\$
Total Mulch Materials Cost/Acre				\$0.00

Application

Description	Cost /Acre
	\$
Total Mulch Application Cost/Acre	\$0.00

NURSERY STOCK PLANTING

Common Name	No / Acre	Type and Size	Planting Cost	Fertilizer Pellet Cost	Cost /Acre
					\$
Totals Nursery Stock Cost / Acre					\$0.00

JOB TIME AND COST

No. of Acres:	8.6	Cost /Acre:	\$338.80
Estimated Failure Rate:	0%	Cost /Acre*:	\$0.00
*Selected Replanting Work Items:	NONE		

Initial Job Cost:	\$2,913.68
Reseeding Job Cost:	\$0.00
Total Job Cost:	\$2,914
Job Hours:	15.00

BOREHOLE SEALING WORK

Task description: Well Abandonment

Site: Lyons Quarry

Permit Action: 2024 Cost Est

Permit/Job#: M1977208

PROJECT IDENTIFICATION

Task #: 036

State: Colorado

Abbreviation: None

Date: 10/22/2024

County: Boulder

Filename: M208-036

User: JPL

Agency or organization name: DRMS

UNIT COSTS

Borehole Description	Sealing/Item Method	Diameter	Length	Quantity	Unit	Unit Cost	Total Cost
4" Wells CEM-001 -005	Portland cement grout (Bag, material cost only...94 lb. bag)	4	542	44.30	bag	\$22.00	\$974.60
2" Wells CEM-002, 003, 004	Portland cement grout (Bag, material cost only...94 lb. bag)	2	190	4.00	bag	\$22.00	\$88.00
1" Piezometers P-001 to 006	Portland cement grout (Bag, material cost only...94 lb. bag)	1	86	1.00	bag	\$22.00	\$22.00
Well Site Reclamation	USER PROVIDED ITEM	1	1	1.00	LS	\$4,747.00	\$4,747.00
Well Crew	General laborer - Colorado (total incl. fringes, empl. burden)	NA	NA	16.00	HR	\$23.44	\$375.04

Job Hours: 16.00

Total Cost: \$6,207.00

EQUIPMENT MOBILIZATION/DEMOBILIZATIONTask description: **Mob/Demob**Site: **Lyons Quarry**Permit Action: **2024 Cost Est**Permit/Job#: **M1977208****PROJECT IDENTIFICATION**Task #: **037**State: **Colorado**Abbreviation: **None**Date: **10/22/2024**County: **Boulder**Filename: **M208-037**User: **JPL**Agency or organization name: **DRMS****EQUIPMENT TRANSPORT RIG COST**Shift basis: **1 per day**Cost Data Source: **CRG Data**Truck Tractor Description: **GENERIC ON-HIGHWAY TRUCK TRACTOR, 6X4, DIESEL POWERED,
400 HP (2ND HALF, 2006)**Truck Trailer Description: **GENERIC FOLDING GOOSENECK, DROP DECK EQUIPMENT
TRAILER (25T, 50T, AND 100T)****Cost Breakdown:**

Available Rig Capacities	0-25 Tons	26-50 Tons	51+ Tons
Ownership Cost/Hour:	\$10.44	\$22.18	\$23.94
Operating Cost/Hour:	\$26.48	\$54.55	\$55.65
Operator Cost/Hour:	\$22.52	\$22.52	\$22.52
Helper Cost/Hour:	\$0.00	\$23.53	\$23.53
Total Unit Cost/Hour:	\$59.44	\$122.78	\$125.64

NON ROADABLE EQUIPMENT:

Machine Description	Weight/ Unit (TONS)	Owner ship Cost/hr/ unit	Haul Rig Cost/hr/unit	Fleet Size	Haul Trip Cost/hr/ fleet	Return Trip Cost/hr/ fleet	DOT Permit Cost/ fleet
Cat D9T - 9SU	66.13	\$271.95	\$125.64	2	\$795.18	\$251.28	\$250.00
CAT 16M	28.73	\$185.14	\$122.78	1	\$307.92	\$122.78	\$250.00
Cat 637G w/push-pull	59.59	\$281.32	\$125.64	6	\$2,441.76	\$753.84	\$500.00
CAT 992K	107.88	\$270.75	\$125.64	2	\$792.78	\$251.28	\$250.00
Cat 777F	80.18	\$199.47	\$125.64	4	\$1,300.44	\$502.56	\$500.00
Drill/Broadcast Seeder with Tractor	25.00	\$41.02	\$59.44	2	\$200.92	\$118.88	\$500.00
Power Mulcher (Bowie LD-90)	6.00	\$27.21	\$59.44	2	\$173.30	\$118.88	\$250.00
Cat 345D L 12'-10" Stick	54.31	\$281.20	\$125.64	1	\$406.84	\$125.64	\$250.00
Water Tanker, 10,000 Gal.	41.10	\$111.67	\$122.78	1	\$234.45	\$122.78	\$250.00
Cat D8T - 8SU	53.08	\$187.85	\$125.64	2	\$626.98	\$251.28	\$500.00
CAT 973D	29.07	\$120.46	\$122.78	11	\$2,675.64	\$1,350.58	\$250.00
Cat 725	24.54	\$119.08	\$59.44	17	\$3,034.84	\$1,010.48	\$750.00
CAT 450E	9.80	\$78.06	\$59.44	1	\$137.50	\$59.44	\$250.00
Grove RT650E, 105', 45.4 MT	28.74	\$189.03	\$122.78	3	\$935.43	\$368.34	\$250.00
Cat 320D L 9'-6" Stick	23.70	\$244.29	\$59.44	4	\$1,214.92	\$237.76	\$250.00
Broderson IC-80-2G, 32', 8.2MT	8.25	\$137.49	\$59.44	1	\$196.93	\$59.44	\$250.00

Cat 315D L 8'-6" Stick	19.05	\$244.29	\$59.44	3	\$911.19	\$178.32	\$750.00
Subtotals:					\$16,387.02	\$5,883.56	\$6,250.00

ROADABLE EQUIPMENT:

Machine Description	Total Cost/hr/ unit	Fleet Size	Haul Trip Cost/hr/ fleet	Return Trip Cost/hr/ fleet
Lube Truck, 4x2, 190 HP	\$41.41	1	\$41.41	\$41.41
Fuel Tanker, 4x2, 170 HP	\$34.10	1	\$34.10	\$34.10
Light Duty Pickup, 4x4, 1 T. Crew	\$24.60	5	\$123.00	\$123.00
Generic 12-18 cy, 6x4	\$114.77	1	\$114.77	\$114.77
Flatbed Truck, 6x4, 45K GVW	\$81.77	1	\$81.77	\$81.77
Subtotals:			\$395.05	\$395.05

EQUIPMENT HAUL DISTANCE and Time

Nearest Major City or Town within project area region:	LONGMONT	
Total one-way travel distance:	12.00	miles
Average Travel Speed:	45.00	mph

Total Non-Roadable Mob/Demob Cost *	\$89,925.72
-------------------------------------	-------------

 ** two round trips with haul rig:

Total Roadable Mob/Demob Cost **	\$210.69
----------------------------------	----------

 ** one round trip, no haul rig:

Transportation Cycle Time:

	Non-Roadable Equipment	Roadable Equipment
Haul Time (Hours):	0.27	0.27
Return Time (Hours):	0.27	0.27
Loading Time (Hours):	1.00	NA
Unloading Time (Hours):	1.00	NA
Subtotals:	2.53	0.53

JOB TIME AND COST

Total job time:	5.07	Hours
-----------------	------	-------

Total job cost:	\$90,136
-----------------	----------